MATH INQUIRERS HOLIDAY PROGRAM

# Question Bank

Core, Think Harder and Challenge questions for every lesson, with answers

**Notice it. Test it. Explain why.**

## Question Bank • Ages 12-13 (finishing P6)

*Math Inquirers P6 Ages 12-13 • Maths + coding bridge. Notice it. Test it. Explain why.*

### How to use this bank

- Each lesson has three levels. Every child starts with Core, then moves up when they are ready. Challenge questions are the ones to talk about together.
- CORE builds fluency with the main idea of the lesson, including reading and tracing short programs. Use it for warm-ups and for children who need confidence.
- THINK HARDER goes a step further: two-step problems, patterns with a hidden rule, unknowns and larger numbers.
- CHALLENGE questions are non-routine. They ask children to find every way, work backwards, test a case, justify an answer or notice what is missing. This is the style of thinking used in gifted-education-type problems. It is not an official GEP standard, and this program is not affiliated with MOE.
- Ask for the reason every time: "How do you know?" A correct answer with no reason is only half done.
- Some questions ask you to predict, trace or plan a short program. Children can test their answer in Scratch (or Python) after they have written their prediction on paper.
- Answers are at the back. Do not hand the answer pages to children.

### What each lesson practises

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| **Lesson 1** | Code the Pattern (Sequences and the nth Term) | 6 + 8 + 9 questions |
| **Lesson 2** | Prime Hunters (Primes, Factors and Prime Factorisation) | 6 + 8 + 9 questions |
| **Lesson 3** | Below Zero (Negative Numbers) | 6 + 8 + 9 questions |
| **Lesson 4** | Algebra Machines (Expressions and Equations) | 6 + 8 + 9 questions |
| **Lesson 5** | Every Possible Way (Counting) | 6 + 8 + 9 questions |
| **Lesson 6** | Fair or Unfair? (Probability) | 6 + 8 + 9 questions |
| **Lesson 7** | Turtle Geometry (Angles in Polygons) | 6 + 8 + 9 questions |
| **Lesson 8** | Secret Codes (Clock (Modular) Arithmetic) | 6 + 8 + 9 questions |
| **Lesson 9** | Data Detectives (Averages and Data) | 6 + 8 + 9 questions |

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| **Lesson 10** | Showcase Challenge (Mixed: Maths and Code) | 6 + 8 + 9 questions |

# Lesson 1: Code the Pattern

*Sequences and the nth Term • Week 1*

## CORE

**C1.** Write the next two terms: 8, 13, 18, 23, …

**C2.** Write the rule for the nth term of 3, 5, 7, 9, …

**C3.** Find the 10th term of 7, 10, 13, 16, …

**C4.** Write the next two terms: 40, 36, 32, 28, …

**C5.** What does this code say? set n to 10, then repeat 3 [change n by 4], then say n.

**C6.** The rule for the nth term is 4n + 3. Write the first three terms.

## THINK HARDER

**T1.** Find the 100th term of 12, 19, 26, 33, …

**T2.** Is 200 a term of 2, 9, 16, 23, …? Explain.

**T3.** Matchstick houses in a row: 1 house uses 6 sticks, 2 houses use 11, 3 houses use 16. How many houses can you make with 100 sticks?

**T4.** What does this code say? set n to 100, then repeat 8 [change n by −7], then say n.

**T5.** The 5th term of a sequence is 23 and the 6th term is 27. The difference is always the same. What is the 1st term? What is the rule?

**T6.** The 1st term of a sequence is 4 and the 10th term is 40. The difference is always the same. What is the difference?

**T7.** Write a sequence whose 1st term is 10 and whose 100th term is 1000.

**T8.** How many terms of 3, 7, 11, 15, … are less than 500?

## CHALLENGE (non-routine: justify, find every way)

**X1.** Pattern 1 is 1 tile, pattern 2 is a 2 by 2 square of tiles, pattern 3 is a 3 by 3 square. How many tiles are in pattern 10? How many more tiles does pattern 11 need than pattern 10?

**X2.** A straight fence is 60 m long. There is a post every 4 m, including one at each end. How many posts are there?

**X3.** The rule for a sequence is 6n − 4. Ben says every term is even. Is he right? Explain.

**X4.** Find the first term of 5, 8, 11, 14, … that is bigger than 1000. Which term is it?

**X5.** A sequence goes 2, 6, 18, 54, … Is the difference the same each time? What is the rule? What is the 7th term?

**X6.** What does this code say? set a to 1, set b to 1, then repeat 5 [change a by b, change b by a], then say b.

**X7.** Ben counts 6, 11, 16, 21, … and Mei counts 83, 81, 79, 77, … They say their terms at the same time. When do they say the same number, and what is it?

**X8.** Which numbers below 60 are in BOTH sequences 2, 5, 8, 11, … and 4, 9, 14, 19, …? Find them all.

**X9.** What does this code say? set t to 0, set k to 0, then repeat 5 [change k by 1, change t by k], then say t. What would it say with repeat 100?

# Lesson 2: Prime Hunters

*Primes, Factors and Prime Factorisation • Week 1*

## CORE

**C1.** List all the factors of 18.

**C2.** Is 37 prime? How do you know?

**C3.** What is 29 mod 4 (the remainder when 29 is divided by 4)?

**C4.** Write 30 as a product of primes.

**C5.** Which of these are prime: 21, 23, 25, 27, 29?

**C6.** Write 20 as the sum of two primes.

## THINK HARDER

**T1.** How many factors does 64 have? List them.

**T2.** Find the smallest number that has exactly 6 factors.

**T3.** Write 100 as the sum of two primes in every possible way.

**T4.** What does this code say? set count to 0, set d to 1, then repeat 45 [if 45 mod d = 0 then change count by 1; change d by 1], then say count.

**T5.** Twin primes differ by 2, like 11 and 13. Find all the twin prime pairs below 50.

**T6.** A number is 2 × 2 × 5 × 7. What is the number? How many factors does it have?

**T7.** Which number below 30 has the most factors?

**T8.** A number between 20 and 40 leaves remainder 1 when divided by 2, by 3 and by 4. Find both possible numbers.

## CHALLENGE (non-routine: justify, find every way)

**X1.** What is the largest prime number below 200?

**X2.** Ali multiplies two different primes and gets 91. Mei multiplies three different primes and gets 105. Which primes did each of them use?

**X3.** Find the smallest number that can be divided exactly by 2, 3, 4, 5 and 6.

**X4.** 1001 = 7 × 11 × 13. Use this to write 3003 and 5005 as products of primes.

**X5.** Mei's program counts how many numbers from 1 to 30 are prime. What number should it say? List them to check.

**X6.** Which even number below 50 can be written as the sum of two primes in the most ways? How many ways?

**X7.** Find all the numbers between 100 and 200 that have exactly 3 factors.

**X8.** A farmer counts her eggs in 2s, 3s, 4s, 5s and 6s, and each time there is 1 egg left over. When she counts them in 7s, none are left over. What is the smallest number of eggs she could have?

**X9.** 20 lockers are closed. Student 1 opens every locker. Student 2 changes every 2nd locker (open to closed, closed to open). Student 3 changes every 3rd locker, and so on up to student 20. Which lockers are open at the end?

# Lesson 3: Below Zero

*Negative Numbers • Week 2*

## CORE

**C1.** Put in order, smallest first: −4, 2, −10, 0, −1.

**C2.** It is −3 °C. The temperature rises by 8 degrees. What is it now?

**C3.** It is 2 °C. The temperature falls by 7 degrees. What is it now?

**C4.** Which is colder, −6 °C or −11 °C? By how many degrees?

**C5.** In a building the floors are numbered ..., −2, −1, 0, 1, 2, ... and 0 is the ground floor. A lift is at floor −2. It goes up 6 floors. Which floor is it on now?

**C6.** A sprite is at x = 30. It runs "change x by −50". Where is it now?

## THINK HARDER

**T1.** Work out −7 + 12 and 5 − 13.

**T2.** A diver is at −14 m. A seagull is flying at 9 m. How far apart are they?

**T3.** Work out 4 − (−6). Hint: how far is it from −6 to 4?

**T4.** Find the missing number: −3 + ____ = −10.

**T5.** At 6 a.m. it was −4 °C. By 2 p.m. it was 7 °C. By 10 p.m. it had fallen 9 degrees. How much did it rise from 6 a.m. to 2 p.m.? What was the temperature at 10 p.m.?

**T6.** set n to 10, then repeat 5 times: change n by −4. Then say n. What does it say?

**T7.** What number is exactly halfway between −9 and 3?

**T8.** A submarine is at −120 m. It rises 45 m, then dives 70 m. Where is it now?

## CHALLENGE (non-routine: justify, find every way)

**X1.** Put −4, −3, −2, −1, 0, 1, 2, 3, 4 in a 3 by 3 grid so that every row, every column and both diagonals add up to 0.

**X2.** At midnight it was −6 °C. It rose 2 degrees every hour until 7 a.m., then 3 degrees every hour. What was the temperature at 10 a.m.?

**X3.** set x to 0, then repeat 6 times: change x by 50, change x by −80. Then say x. What does it say?

**X4.** Which is bigger, −3 − 4 or −3 − (−4)? By how much?

**X5.** Tokyo is 6 °C. Seoul is 4 degrees colder than Tokyo. Moscow is 17 degrees colder than Seoul. What is the temperature in Moscow?

**X6.** Lift A starts at the surface of a mine (0 m) and goes down 35 m every minute. Lift B starts at −300 m and goes up 25 m every minute. When are they at the same height, and at what height?

**X7.** A sprite starts at x = −10. It moves +5, then −3, then +5, then −3, and so on. After how many moves is it first at a number greater than 0? Where is it?

**X8.** Two numbers add up to −2. Their difference is 10. What are the two numbers?

**X9.** In many Singapore buildings there is no level 0: level 1 is the ground floor and B1 is the level below it. A lift goes from level 4 down to B3. How many floors does it go down? (Count the levels one by one; do not use a number line with 0.)

# Lesson 4: Algebra Machines

*Expressions and Equations • Week 2*

## CORE

**C1.** The rule is OUT = 4n + 1. What is OUT when n = 6?

**C2.** IN 1, 2, 3, 4 gives OUT 3, 5, 7, 9. Write the rule with n.

**C3.** Solve 3n = 54.

**C4.** A machine multiplies by 6, then takes away 4. OUT is 38. What was IN?

**C5.** Ali has n stickers. Mei has 5 more than Ali. Write how many they have altogether.

**C6.** A Scratch machine does: set out to (answer * 3) + 4, then say out. You type 7. What does it say?

## THINK HARDER

**T1.** IN 1, 2, 3, 4 gives OUT 1, 7, 13, 19. Find the rule. What is OUT when IN is 20?

**T2.** Solve 9n − 7 = 65.

**T3.** Machine A does × 2, then + 10. Machine B does + 5, then × 2. Put 7 into both. What do you notice? Explain with n.

**T4.** I think of a number, divide it by 4, then add 6. I get 15. What is my number?

**T5.** IN 3 gives OUT 11. IN 7 gives OUT 23. The rule is "× a number, then + a number". Find it.

**T6.** set n to 2, then repeat 3 times: set n to (n × 2) + 1. Then say n. What does it say?

**T7.** Solve (n + 6) × 4 = 60.

**T8.** A taxi charges $4 to start, plus $2 for every km. Write the cost in dollars for n km. How far can you go for $30?

## CHALLENGE (non-routine: justify, find every way)

**X1.** A machine does × 3, then − 4. Which IN number comes out as the SAME number?

**X2.** Two machines are joined: the OUT of the first goes into the second. Each one does × 2, then + 1. IN 5 gives OUT 23. Which IN gives OUT 63?

**X3.** A rule is "× a number, then + a number". IN 2 gives OUT 10 and IN 6 gives OUT 22. What is OUT when IN is 0?

**X4.** set n to 0, then repeat until (n × 7) + 3 > 100: change n by 1. Then say n. What does it say?

**X5.** Machine M does × 5, then − 3. Machine N does × 2, then + 15. Which IN gives the same OUT on both machines? What is that OUT?

**X6.** Which is bigger, 3n + 5 or 3(n + 2)? Does it depend on n?

**X7.** Three numbers in a row add up to 87. Call the smallest n. Write the sum with n and find the numbers.

**X8.** Mei is n years old. Her mother is 3 times as old. In 12 years, her mother will be twice as old as Mei. How old is Mei now?

**X9.** Ali's secret machine does × a number, then + or − a number. IN 4 gives OUT 9. IN 9 gives OUT 24. What does IN 20 give? Which IN gives OUT 99?

# Lesson 5: Every Possible Way

*Counting • Week 3*

## CORE

**C1.** A shop sells 3 shirts and 5 pairs of trousers. How many different outfits of 1 shirt and 1 pair of trousers can you make?

**C2.** List every 2-digit number you can make from the cards 4, 6 and 9, using each card at most once.

**C3.** 4 friends meet. Every pair shakes hands once. How many handshakes?

**C4.** How many times does this say “hi”? repeat 4 { repeat 6 { say hi } }

**C5.** A grid is 3 squares wide and 1 square tall. How many routes go from the bottom left corner to the top right corner, going only right or up?

**C6.** A lock has 2 dials. Each dial shows a digit from 0 to 9. How many different codes are there?

## THINK HARDER

**T1.** A set lunch is 1 main, 1 drink and 1 dessert. There are 4 mains and 3 drinks. There are 60 different set lunches. How many desserts are there?

**T2.** How many 3-digit numbers can you make from the cards 1, 2, 3 and 4, using each card at most once? How many of them are even?

**T3.** In a league of 6 teams, every team plays every other team twice (once at home, once away). How many matches are played?

**T4.** A 4-digit PIN uses only the digits 1, 2 and 3. Digits may repeat. How many PINs are there?

**T5.** A hexagon has 6 corners. How many diagonals does it have?

**T6.** How many rectangles of any size can you find in a strip of 4 squares in a row?

**T7.** 5 children line up for a photo. In how many different orders can they stand? In how many of these is Ali at the front?

**T8.** How many routes go from one corner of a grid 4 squares wide and 2 squares tall to the opposite corner, going only right or up?

## CHALLENGE (non-routine: justify, find every way)

**X1.** How many different triangles can you make by joining 3 of the 5 corners of a pentagon?

**X2.** How many 2-digit numbers have two different digits?

**X3.** How many digits are printed to number the pages of a book from page 1 to page 120?

**X4.** An ice cream shop has 6 flavours. A cup has 2 different flavours, and the order does not matter. How many different cups are there?

**X5.** 4 friends each send a card to every other friend. How many cards are sent?

**X6.** How many squares of any size are there on a 3 by 3 grid of squares?

**X7.** How many routes go from one corner of a 4 by 4 grid of squares to the opposite corner, going only right or up?

**X8.** A set of dominoes has one domino for every pair of spot numbers from 0 to 6, including doubles such as 3 | 3. How many dominoes are in the set?

**X9.** How many 3-digit numbers have digits that add up to 5? (For example, 203.)

# Lesson 6: Fair or Unfair?

*Probability • Week 3*

## CORE

**C1.** Roll a dice. What is the probability of rolling a 4?

**C2.** Roll a dice. What is the probability of rolling an even number?

**C3.** A bag has 3 red and 7 yellow counters. You pick one without looking. What is the probability it is yellow?

**C4.** You flip a coin 200 times. About how many heads do you expect?

**C5.** A spinner has 4 equal parts: red, red, blue, green. What is the probability of red?

**C6.** Roll a normal dice. What is the probability of rolling a 9?

## THINK HARDER

**T1.** A spinner has 10 equal parts numbered 1 to 10. What is the probability of spinning a prime number?

**T2.** Roll two dice and add. What is the probability that the total is 9?

**T3.** A bag has 5 red counters and some blue counters. The probability of picking red is 1/4. How many blue counters are there?

**T4.** Ali flips a coin 3 times. What is the probability that he gets heads all 3 times?

**T5.** One letter is picked at random from the word MATHEMATICS. What is the probability that it is M?

**T6.** A program rolls a dice 1200 times. About how many even numbers should it roll?

**T7.** Roll two dice. What is the probability that at least one of them shows a 6?

**T8.** Roll a dice. Ali wins on 1 or 2; Ben wins on 3, 4, 5 or 6. Ben scores 1 point for each win. How many points should Ali score for each win to make the game fair?

## CHALLENGE (non-routine: justify, find every way)

**X1.** A bag has red and blue counters in the ratio 2 : 3. What is the probability of picking red?

**X2.** A number from 1 to 30 is picked at random. What is the probability that it is a multiple of 3 or a multiple of 5?

**X3.** Roll two dice and multiply the numbers. What is the probability that the answer is odd?

**X4.** A spinner is spun 1000 times in a program and lands on red 248 times. Is red most likely 1/2, 1/4 or 1/8 of the spinner? Explain.

**X5.** A bag has 4 red and 6 blue counters. Ali picks a red counter and keeps it. What is the probability that the next counter picked is red?

**X6.** Ben says: “When you flip two coins you get two heads, two tails, or one of each, so the chance of one of each is 1/3.” Is he right?

**X7.** Three coins are flipped. What is the probability of getting exactly 2 heads?

**X8.** Mei and Ali each roll one dice. Mei wins if her number is bigger. Ali wins if his number is bigger or the same. Is the game fair? What is the probability that Mei wins?

**X9.** Spinner A has 3 equal parts: 1, 2, 3. Spinner B has 4 equal parts: 1, 2, 3, 4. Spin both and multiply. Even wins if the answer is even, Odd wins if it is odd. Is the game fair?

# Lesson 7: Turtle Geometry

*Angles in Polygons • Week 4*

## CORE

**C1.** A turtle draws a regular hexagon. How many degrees does it turn at each corner?

**C2.** What is each inside angle of a regular hexagon?

**C3.** Fill in the turn so the turtle draws a square: repeat 4 [move 100, turn ↻ ___ ]

**C4.** A regular polygon has a turn angle of 72°. How many sides does it have?

**C5.** At a corner the turtle turns 50°. What is the inside angle at that corner?

**C6.** A turtle draws any triangle and comes back to the start facing the same way. How many degrees has it turned altogether?

## THINK HARDER

**T1.** Each inside angle of a regular polygon is 4 times its turn angle. How many sides does the polygon have?

**T2.** Each inside angle of a regular polygon is 140°. What do all its inside angles add up to?

**T3.** Mei's turtle draws a shape with 4 sides. Its first three turns are 70°, 100° and 90°. It comes home facing the same way after going round once. What is its last turn? What is the inside angle at that corner?

**T4.** A regular polygon has n sides. Write its turn angle and its inside angle using n.

**T5.** In a regular polygon the turn angle is the same as the inside angle. Which polygon is it?

**T6.** Which regular polygons with 10 to 20 sides have a whole-number turn angle?

**T7.** Regular hexagons fit together around a point with no gaps, like a honeycomb. Explain why using angles. Can regular pentagons do the same?

**T8.** A turtle draws a regular hexagon, then a square, then an equilateral triangle, each once round, ending each shape facing the way it started. How many degrees does it turn altogether?

## CHALLENGE (non-routine: justify, find every way)

**X1.** Ali wants a regular pentagon. He types the inside angle by mistake: repeat 5 [move 100, turn ↻ 108]. Does his shape close? How many repeats would make it close, and what would it draw?

**X2.** How many different stars with 9 points can a turtle draw using equal moves and equal turns? Give the turns.

**X3.** Each inside angle of a regular polygon is 5 times its turn angle. How many sides does it have?

**X4.** How many different 12-point stars can a turtle draw with equal moves and equal turns? Give the turn and how many times it goes round.

**X5.** A turtle runs: repeat 12 [repeat 4 (move 100, turn ↻ 90), turn ↻ 30]. How many full turns does it make altogether?

**X6.** Each inside angle of a regular polygon is 20° bigger than each inside angle of a regular hexagon. How many sides does it have?

**X7.** All the inside angles of a regular polygon add up to 1800°. How many sides does it have, and what is its turn angle?

**X8.** A square, a hexagon and a 12-sided shape fit exactly around a point: 90° + 120° + 150° = 360°. Find one other set of three regular polygons whose inside angles add up to 360° (hint: the answer uses a square or a triangle).

**X9.** Try turn 120 and turn 60 with repeat 6 and move 100. Which one draws a hexagon? Why does the other not make a 6-point star?

# Lesson 8: Secret Codes

*Clock (Modular) Arithmetic • Week 4*

## CORE

**C1.** It is 10 o'clock. What time will it be 5 hours later?

**C2.** Work out 20 mod 12.

**C3.** What is the remainder when 45 is divided by 7?

**C4.** Today is Monday. What day will it be 10 days from today?

**C5.** Code the word DOG with a shift of 1.

**C6.** Decode KBN. It was made with a shift of 1.

## THINK HARDER

**T1.** The pattern CODECODECODE... goes on and on. What is the 100th letter?

**T2.** It is 10 o'clock now. What time will it be 1000 hours from now (on a 12-hour clock)?

**T3.** Find the smallest number bigger than 10 that leaves a remainder of 3 when it is divided by 4 and also when it is divided by 5.

**T4.** Decode BYFFI. It was made with a shift of 20.

**T5.** Today is Monday. What day will it be in 365 days? In 366 days?

**T6.** Mei codes with a shift of 3. Ben codes with a shift of 29. Do their codes give the same answer? Why?

**T7.** Work out 47 mod 12, 35 mod 12 and (47 + 35) mod 12. What do you notice?

**T8.** On a 24-hour clock it is 21:00. What time will it be 50 hours later?

## CHALLENGE (non-routine: justify, find every way)

**X1.** Mei makes a new code: multiply each letter number by 2, then take mod 26. Why can her friend not always decode it?

**X2.** Ben's code multiplies each letter number by 3, then takes mod 26. Code the word CAB. Do two letters ever get the same code letter?

**X3.** What day of the week is it 1000 days after a Monday?

**X4.** Find the smallest number that leaves a remainder of 1 when it is divided by 2, 3, 4, 5 or 6, and is a multiple of 7.

**X5.** Count 1, 2, 3, 4, 5, 6, 7, 1, 2, 3, ... (start again after 7). What is the 2026th number you say?

**X6.** Ali codes a message with a shift of k, then codes the result again with the same shift k. It is the same as one shift of 6. What could k be? Find every answer from 0 to 25.

**X7.** Crack this Caesar code. Hint: a one-letter word is usually A or I. X ADKT BPIWH

**X8.** Ali codes a message with a shift of 7. Mei codes Ali's result with another shift. The final message is the same as the original coded with a shift of 2. What shift did Mei use?

**X9.** Start at 12 on a clock and keep jumping forward 5 hours. Do you land on every number before you get back to 12? What about jumps of 4 hours?

# Lesson 9: Data Detectives

*Averages and Data • Week 5*

## CORE

**C1.** Find the mean of 3, 5, 7 and 9.

**C2.** Find the mean of 10, 0, 8 and 6.

**C3.** The mean of 5 numbers is 11. What is their total?

**C4.** Four bags have a mean mass of 2.5 kg. What is their total mass?

**C5.** Find the median and the mode of 4, 9, 2, 9, 6.

**C6.** A Scratch list called marks holds 6, 8 and 7. A loop starts total at 0 and adds every item to total. What is total after the loop? What is the mean?

## THINK HARDER

**T1.** The mean height of 4 children is 150 cm. A fifth child joins the group and the mean height becomes 152 cm. How tall is the fifth child?

**T2.** Mei's mean score for 3 tests is 84. What must she score in the 4th test to raise her mean to 86?

**T3.** Write 5 whole numbers with a mean of 6 and a mode of 4.

**T4.** A Scratch list holds 8 numbers with a mean of 12. Then the number 21 is added to the list. What will the mean program say now?

**T5.** The mean of 7, 11, x and 14 is 10. What is x?

**T6.** The mean of five numbers is 20. Each number is doubled and then 3 is taken away. What is the new mean?

**T7.** Ali's mean for 5 spelling tests is 16. His mean for the first 3 tests is 15. What is the mean of his last 2 tests?

**T8.** Five whole numbers have a mean of 9, a median of 8 and a mode of 8. The smallest is 5 and the largest is 14. Find all five numbers.

## CHALLENGE (non-routine: justify, find every way)

**X1.** Five numbers have a mean of 8. One number is removed, and the mean of the other four is 7. Which number was removed?

**X2.** A basketball player's mean score for 9 games is 30 points. How many points must she score in the 10th game to raise her mean to 32?

**X3.** Five whole numbers have a mean of 6, a median of 5 and only one mode, 2. What is the largest the biggest number can be?

**X4.** Can the mean of 5 whole numbers be bigger than 4 of them? Give an example.

**X5.** The mean of 4 numbers is 15. One of the numbers is changed and the mean becomes 18. By how much was that number changed?

**X6.** Jun's program adds 5 to every item in a list of 10 numbers, then doubles every item. The mean was 7 before. What is the mean now?

**X7.** Find ALL the sets of three different counting numbers whose mean is 4 and whose median is 4.

**X8.** The mean of 1, 2, 3, … , 99 is 50. Explain why without adding them all up. Then plan a short program that checks it.

**X9.** A group of 8 children has a mean mass of 41 kg. When their tutor joins, the mean mass of all 9 becomes 44 kg. What is the tutor's mass?

# Lesson 10: Showcase Challenge

*Mixed: Maths and Code • Week 5*

## CORE

**C1.** Find the 20th term of 5, 8, 11, 14, ...

**C2.** Write 60 as a product of prime numbers.

**C3.** The temperature is −3 °C. It falls by 5 °C. What is it now?

**C4.** A machine does × 4, then − 6. What comes out when 5 goes in?

**C5.** How many different 2-digit numbers can you make with the digit cards 3, 5 and 7 (each card used at most once)?

**C6.** Find the mean of 12, 15, 9, 20 and 14.

## THINK HARDER

**T1.** A row of squares uses 4, 7, 10, 13, ... matchsticks. Which pattern number uses exactly 100 matchsticks?

**T2.** Find all the prime numbers between 80 and 100.

**T3.** A regular polygon has an angle of 150° inside each corner. How many sides does it have?

**T4.** Two dice are rolled and the numbers are added. What is the probability that the total is 10 or more?

**T5.** Use a Caesar cipher with a shift of 5 to encode the word MATHS.

**T6.** It is 10 o'clock. What time will a 12-hour clock show 50 hours later?

**T7.** The mean of 6 numbers is 9. One more number is added and the mean becomes 10. What number was added?

**T8.** In how many different orders can 4 children stand in a line?

## CHALLENGE (non-routine: justify, find every way)

**X1.** How many of the numbers from 1 to 100 are NOT multiples of 2 or of 3?

**X2.** A machine does × 3, then − 8. Mei puts a number in and gets the SAME number out. What is it? What if the machine does × 3, then + 8?

**X3.** How many squares of any size are there on a 4 by 4 grid?

**X4.** What is the smallest number with exactly 6 factors?

**X5.** A turtle draws a star: repeat 5 times (move 100, turn 144°). How many full turns does the turtle make altogether?

**X6.** Three numbers have a mean of 8 and a median of 7. The biggest is 9 more than the smallest. Find the three numbers.

**X7.** Find ALL the ways to write 50 as the sum of two prime numbers.

**X8.** A spinner has the numbers 1 to 8, all equally likely. Ravi spins it twice and adds the numbers. What is the probability that the total is 9? Plan a program that tests it 1000 times.

**X9.** The nth term of a sequence is 6n − 1. Is 2027 (the year you start Secondary 1) in the sequence? Which is the first term bigger than 2027?

# Answer key (tutor only)

## Lesson 1: Code the Pattern

**C1.** 28, 33 (add 5)

**C2.** 2n + 1

**C3.** 34 (rule 3n + 4: 3 × 10 + 4)

**C4.** 24, 20 (take away 4)

**C5.** 22 (10 + 4 + 4 + 4)

**C6.** 7, 11, 15

**T1.** 705. The rule is 7n + 5, and 7 × 100 + 5 = 705.

**T2.** No. The rule is 7n − 5. The 29th term is 198 and the 30th term is 205, so 200 is skipped.

**T3.** 19 houses. The rule is 5n + 1: 19 houses use 96 sticks (4 left over) and 20 houses would need 101.

**T4.** 44 (100 − 8 × 7 = 100 − 56)

**T5.** 7. Go back 4 jumps of 4: 23 − 16 = 7. The rule is 4n + 3.

**T6.** 4. From the 1st term to the 10th term is 9 jumps, and 36 ÷ 9 = 4. (The rule is 4n.)

**T7.** 10, 20, 30, 40, … (rule 10n). There are 99 jumps, and 990 ÷ 99 = 10.

**T8.** 125. The rule is 4n − 1. The 125th term is 499 and the 126th is 503.

**X1.** 100 tiles. Pattern 11 has 121 tiles, so 21 more. The differences are 3, 5, 7, 9, …, so this is NOT a constant-difference sequence.

**X2.** 16 posts. 60 ÷ 4 = 15 gaps, and there is one more post than gaps.

**X3.** Yes. 6 × n is always even, and an even number take away 4 is even (2, 8, 14, 20, …).

**X4.** 1001, the 333rd term. (3n + 2: the 332nd term is 998 and the 333rd is 1001.)

**X5.** No: the differences are 4, 12, 36. Each term is 3 times the one before. The 7th term is 1458 (54, 162, 486, 1458).

**X6.** 144. a and b leapfrog: 2, 3, 5, 8, 13, 21, 34, 55, 89, 144 (the Fibonacci numbers).

**X7.** At the 12th term they both say 61. The gap starts at 77 and closes by 7 each time (Ben +5, Mei −2): 77 ÷ 7 = 11 jumps after the 1st term. Check: 5 × 12 + 1 = 61 and 85 − 2 × 12 = 61.

**X8.** 14, 29, 44 and 59. One more than each of them is a multiple of both 3 and 5, so a multiple of 15 (15, 30, 45, 60).

**X9.** 15 (1 + 2 + 3 + 4 + 5). With repeat 100 it says 5050: pair 1 + 100, 2 + 99, … to make 50 pairs of 101.

## Lesson 2: Prime Hunters

**C1.** 1, 2, 3, 6, 9, 18

**C2.** Yes. Its only factors are 1 and 37 (2, 3, 5 do not divide it, and 7 × 7 = 49 is already bigger than 37).

**C3.** 1 (4 × 7 = 28)

**C4.** 2 × 3 × 5

**C5.** 23 and 29

**C6.** 3 + 17 or 7 + 13

**T1.** 7: 1, 2, 4, 8, 16, 32, 64 (8 × 8 counts once).

**T2.** 12 (1, 2, 3, 4, 6, 12). Every number below 12 has fewer than 6 factors.

**T3.** 6 ways: 3 + 97, 11 + 89, 17 + 83, 29 + 71, 41 + 59, 47 + 53.

**T4.** 6. It counts the factors of 45: 1, 3, 5, 9, 15, 45.

**T5.** 6 pairs: 3 and 5, 5 and 7, 11 and 13, 17 and 19, 29 and 31, 41 and 43.

**T6.** 140. It has 12 factors: 1, 2, 4, 5, 7, 10, 14, 20, 28, 35, 70, 140.

**T7.** 24, with 8 factors (1, 2, 3, 4, 6, 8, 12, 24).

**T8.** 25 and 37. Each is 1 more than a multiple of 12.

**X1.** 199. Test 2, 3, 5, 7, 11 and 13 (14 × 14 = 196, 15 × 15 = 225): none of them divides 199.

**X2.** Ali: 7 × 13. Mei: 3 × 5 × 7.

**X3.** 60. It needs 2 × 2 (for 4), a 3 and a 5: 2 × 2 × 3 × 5 = 60.

**X4.** 3003 = 3 × 7 × 11 × 13 and 5005 = 5 × 7 × 11 × 13.

**X5.** 10: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29.

**X6.** 48, in 5 ways: 5 + 43, 7 + 41, 11 + 37, 17 + 31, 19 + 29.

**X7.** 121 and 169 (11 × 11 and 13 × 13). 144 = 12 × 12 does not work because 12 is not prime.

**X8.** 301. It must be 1 more than a multiple of 60: 61, 121, 181, 241, 301, … The first of these in the 7 times table is 301 = 7 × 43.

**X9.** 1, 4, 9 and 16. Locker n is changed once for each factor of n. Only square numbers have an odd number of factors, so only they end up open.

## Lesson 3: Below Zero

**C1.** −10, −4, −1, 0, 2

**C2.** 5 °C

**C3.** −5 °C

**C4.** −11 °C, by 5 degrees

**C5.** Floor 4 (−2 + 6 = 4)

**C6.** x = −20

**T1.** 5 and −8

**T2.** 23 m (14 up to sea level, then 9 more)

**T3.** 10

**T4.** −7 (adding −7 moves 7 to the left)

**T5.** It rose 11 degrees; at 10 p.m. it was −2 °C (7 − 9).

**T6.** −10 (10 − 20)

**T7.** −3 (the gap is 12; −9 + 6 = −3 and 3 − 6 = −3)

**T8.** −145 m (−120 + 45 = −75, −75 − 70 = −145)

**X1.** One answer: top row −3, 2, 1; middle row 4, 0, −4; bottom row −1, −2, 3. 0 must be in the middle and 4 and −4 cannot be in corners; the other 7 answers are turns and flips of this one.

**X2.** 17 °C (−6 + 7 × 2 = 8 at 7 a.m., then 8 + 3 × 3 = 17)

**X3.** −180 (each repeat moves −30; 6 × −30 = −180)

**X4.** −3 − (−4) = 1 is bigger than −3 − 4 = −7, by 8.

**X5.** −15 °C (Seoul 2 °C, then 2 − 17 = −15)

**X6.** After 5 minutes, at −175 m (the gap of 300 m closes by 60 m each minute).

**X7.** After 7 moves, at x = 1 (−5, −8, −3, −6, −1, −4, 1).

**X8.** 4 and −6 (4 + (−6) = −2 and 4 − (−6) = 10)

**X9.** 6 floors (4, 3, 2, 1, B1, B2, B3: six moves down). A number line with 0 would give 7, because it counts a level 0 that these buildings do not have.

## Lesson 4: Algebra Machines

**C1.** 25

**C2.** OUT = 2n + 1

**C3.** n = 18

**C4.** 7 (38 + 4 = 42, 42 ÷ 6 = 7)

**C5.** 2n + 5 (n + n + 5)

**C6.** 25

**T1.** OUT = 6n − 5; IN 20 gives 115.

**T2.** n = 8 (65 + 7 = 72, 72 ÷ 9 = 8)

**T3.** Both give 24. (n + 5) × 2 = 2n + 10, so they always give the same OUT.

**T4.** 36 (15 − 6 = 9, 9 × 4 = 36)

**T5.** OUT = 3n + 2 (IN up 4, OUT up 12, so × 3; 9 + 2 = 11)

**T6.** 23 (2 → 5 → 11 → 23)

**T7.** n = 9 (60 ÷ 4 = 15, 15 − 6 = 9)

**T8.** 2n + 4 dollars; 13 km (30 − 4 = 26, 26 ÷ 2 = 13)

**X1.** 2 (2 × 3 − 4 = 2; 3n − 4 = n)

**X2.** 15 (63 → 31 → 15; check 15 → 31 → 63)

**X3.** 4 (the rule is 3n + 4)

**X4.** 14 (13 gives 94, 14 gives 101)

**X5.** IN 6, OUT 27 (5n − 3 = 2n + 15, so 3n = 18)
**X6.** 3(n + 2) = 3n + 6 is always 1 bigger; it does not depend on n.
**X7.** n + (n + 1) + (n + 2) = 3n + 3 = 87, so n = 28: the numbers are 28, 29, 30.
**X8.** 12 (3n + 12 = 2(n + 12), so n = 12; mother 36; in 12 years 24 and 48)
**X9.** The rule is 3n − 3. IN 20 gives 57. OUT 99 comes from IN 34.

## Lesson 5: Every Possible Way

**C1.** 15 (3 × 5)
**C2.** 6 numbers: 46, 49, 64, 69, 94, 96
**C3.** 6 (3 + 2 + 1)
**C4.** 24 (4 × 6)
**C5.** 4 (go up at one of the 4 lines across)
**C6.** 100 (10 × 10: 00 to 99)
**T1.** 5 (4 × 3 = 12 and 60 ÷ 12 = 5)
**T2.** 24 (4 × 3 × 2); 12 are even (the last card is 2 or 4: 2 choices, then 3 × 2 for the others)
**T3.** 30 (6 × 5: each ordered pair is one home match)
**T4.** 81 (3 × 3 × 3 × 3)
**T5.** 9 (6 corners make 6 × 5 ÷ 2 = 15 lines; 6 of them are sides, so 9 are diagonals)
**T6.** 10 (4 of length 1, 3 of length 2, 2 of length 3, 1 of length 4)
**T7.** 120 (5 × 4 × 3 × 2 × 1); 24 with Ali at the front (4 × 3 × 2 × 1)
**T8.** 15
**X1.** 10
**X2.** 81 (90 two-digit numbers, minus the 9 with both digits the same: 11, 22, ..., 99)
**X3.** 252 (9 one-digit + 90 × 2 + 21 × 3 = 9 + 180 + 63)
**X4.** 15 (6 × 5 ÷ 2)
**X5.** 12 (4 × 3: here A to B and B to A are different cards)
**X6.** 14 (9 small, 4 of size 2 by 2, 1 of size 3 by 3)
**X7.** 70 (add as you go: the corner numbers on the diagonal row are 1, 4, 10, 20, 35, then 70)
**X8.** 28 (7 doubles, plus 6 + 5 + 4 + 3 + 2 + 1 = 21 with two different numbers)
**X9.** 15 (first digit 1: 5 numbers, 2: 4, 3: 3, 4: 2, 5: 1; they are 104, 113, 122, 131, 140, 203, 212, 221, 230, 302, 311, 320, 401, 410, 500)

## Lesson 6: Fair or Unfair?

**C1.** 1/6
**C2.** 3/6 = 1/2 (2, 4 or 6)
**C3.** 7/10
**C4.** About 100 (1/2 × 200), probably not exactly 100
**C5.** 2/4 = 1/2
**C6.** 0: it is impossible
**T1.** 4/10 = 2/5 (2, 3, 5, 7)
**T2.** 4/36 = 1/9 (3+6, 4+5, 5+4, 6+3)
**T3.** 15 (5 out of 20 is 1/4)
**T4.** 1/8 (8 equally likely outcomes: HHH, HHT, HTH, HTT, THH, THT, TTH, TTT)
**T5.** 2/11
**T6.** About 600 (1/2 × 1200)
**T7.** 11/36 (6 with the red die, 6 with the blue die, minus the 6 + 6 counted twice: 6 + 6 − 1 = 11)
**T8.** 2 points (Ali wins 1/3 of the time and Ben 2/3, so Ali needs twice as many points per win)
**X1.** 2/5
**X2.** 14/30 = 7/15 (10 multiples of 3, 6 of 5, and 15 and 30 are counted twice: 10 + 6 − 2 = 14)
**X3.** 9/36 = 1/4 (both dice must be odd: 3 × 3 = 9 ways)
**X4.** 1/4: we would expect about 250 reds, and 248 is close. 1/2 would give about 500 and 1/8 about 125.
**X5.** 3/9 = 1/3 (3 red left out of 9)

**X6.** No: HH, HT, TH, TT are the 4 equally likely outcomes, and one of each happens 2 ways: 2/4 = 1/2.

**X7.** 3/8 (HHT, HTH, THH out of 8 outcomes)

**X8.** Not fair. Mei wins 15/36 = 5/12; Ali wins 21/36 = 7/12 (15 bigger + 6 the same). Fair version: the same number means roll again.

**X9.** Not fair. There are 3 × 4 = 12 outcomes; the answer is odd only when both are odd: 2 × 2 = 4 ways. P(Odd) = 4/12 = 1/3, P(Even) = 8/12 = 2/3.

## Lesson 7: Turtle Geometry

**C1.** 60° (360 ÷ 6)

**C2.** 120° (180 – 60)

**C3.** 90

**C4.** 5 (360 ÷ 72): a regular pentagon

**C5.** 130° (180 – 50)

**C6.** 360°

**T1.** 10 (turn + 4 × turn = 180°, so the turn is 36° and 360 ÷ 36 = 10; check: inside 144° = 4 × 36°)

**T2.** 1260° (turn 40°, so 9 sides; 9 × 140° = 1260°; check (9 – 2) × 180° = 1260°)

**T3.** Last turn 100° (360 – 70 – 100 – 90); inside angle 80° (180 – 100)

**T4.** Turn = 360 ÷ n; inside angle = 180 – 360 ÷ n

**T5.** A square (turn = inside = 90°, because they add up to 180°)

**T6.** 10, 12, 15, 18 and 20 sides (turns 36°, 30°, 24°, 20°, 18°); 360 cannot be shared exactly into 11, 13, 14, 16, 17 or 19 equal turns

**T7.** Three hexagon angles make 3 × 120° = 360°. Pentagons cannot: 3 × 108° = 324° leaves a gap and 4 × 108° = 432° overlaps.

**T8.** 1080° (360° for each shape: 3 × 360°)

**X1.** No: 5 × 108° = 540° is not a whole number of turns. 108 × 10 = 1080 = 3 × 360, so 10 repeats close it: a 10-point star that goes round 3 times (each point 72°).

**X2.** 2: turn 80° (round 2 times) and turn 160° (round 4 times). Turn 120° (3 times round) only draws a triangle 3 times; turns of 200°, 280° … give the same stars turned the other way.

**X3.** 12 (6 × turn = 180°, turn 30°, 360 ÷ 30 = 12)

**X4.** Only 1: turn 150°, going round 5 times (5 × 360 ÷ 12). Turns 60°, 90° and 120° (round 2, 3, 4 times) just redraw a hexagon, square or triangle, and 180° goes back and forth on a line.

**X5.** 13 (12 × (360° + 30°) = 4680° = 13 × 360°)

**X6.** 9 (inside 140°, turn 40°, 360 ÷ 40 = 9)

**X7.** 12 sides ((12 – 2) × 180° = 1800°); turn 30°

**X8.** Square + octagon + octagon (90° + 135° + 135°) or triangle + 12-gon + 12-gon (60° + 150° + 150°).

**X9.** Turn 60 draws the hexagon (6 × 60° = 360°, one full turn). Turn 120 draws a triangle twice round (6 × 120° = 720°, two full turns), and it only visits 3 corners, not 6.

## Lesson 8: Secret Codes

**C1.** 3 o'clock (15 – 12 = 3)

**C2.** 8 (20 = 1 × 12 + 8)

**C3.** 3 (45 = 6 × 7 + 3)

**C4.** Thursday (10 mod 7 = 3; Monday + 3 days)

**C5.** EPH

**C6.** JAM

**T1.** E (100 mod 4 = 0, so the 100th letter is the 4th letter of CODE)

**T2.** 2 o'clock (1000 mod 12 = 4, and 10 + 4 = 14, 14 – 12 = 2)

**T3.** 23 (3 more than a multiple of 20: 23 = 5 × 4 + 3 = 4 × 5 + 3)

**T4.** HELLO (B = 1: 1 – 20 = –19, + 26 = 7 = H; Y = 24 – 20 = 4 = E; F = 5 – 20 = –15, + 26 = 11 = L; I = 8 – 20 = –12, + 26 = 14 = O)

**T5.** Tuesday and Wednesday (365 = 52 × 7 + 1, 366 = 52 × 7 + 2)

**T6.** Yes: 29 mod 26 = 3, so a shift of 29 is a full turn (26) plus 3.

**T7.** 11, 11 and 10. Adding the remainders and taking mod 12 again gives the same answer: (11 + 11) mod 12 = 22 mod 12 = 10.

**T8.** 23:00 (50 mod 24 = 2, and 21 + 2 = 23)

**X1.** Two letters can give the same code letter: A (0) and N (13) both give 0 = A (13 × 2 = 26). So every code letter could stand for two different letters.

**X2.** GAD (C = 2 gives 6 = G, A = 0 gives 0 = A, B = 1 gives 3 = D). No: all 26 letters get different code letters, because 3 and 26 have no common factor.

**X3.** Sunday (1000 = 142 × 7 + 6, Monday + 6 days)

**X4.** 301 (it is 1 more than a multiple of 60: 61, 121, 181, 241, 301; only 301 = 7 × 43)

**X5.** 3 (2026 = 289 × 7 + 3)

**X6.** 3 or 16 (3 + 3 = 6; 16 + 16 = 32 and 32 mod 26 = 6)

**X7.** I LOVE MATHS (X = 23 is I = 8, so the shift is 15)

**X8.** 21 (7 + 21 = 28, and 28 mod 26 = 2)

**X9.** Jumps of 5: yes, all 12 numbers (5, 10, 3, 8, 1, 6, 11, 4, 9, 2, 7, 12). Jumps of 4: no, only 4, 8 and 12, because 4 shares a factor with 12.

## Lesson 9: Data Detectives

**C1.** 6 (24 ÷ 4)

**C2.** 6 (24 ÷ 4: the 0 counts as a value)

**C3.** 55

**C4.** 10 kg

**C5.** In order: 2, 4, 6, 9, 9. Median 6, mode 9.

**C6.** Total 21, mean 7

**T1.** 160 cm (5 × 152 = 760, 4 × 150 = 600, 760 − 600 = 160)

**T2.** 92 (4 × 86 = 344, 3 × 84 = 252, 344 − 252 = 92)

**T3.** Open, e.g. 4, 4, 5, 8, 9 (total 30, and 4 appears most often). Check the total is 30.

**T4.** 13 (8 × 12 = 96, 96 + 21 = 117, 117 ÷ 9 = 13)

**T5.** 8 (4 × 10 = 40, 40 − 7 − 11 − 14 = 8)

**T6.** 37 (20 × 2 − 3)

**T7.** 17.5 (5 × 16 = 80, 3 × 15 = 45, 80 − 45 = 35, 35 ÷ 2 = 17.5)

**T8.** 5, 8, 8, 10, 14 (total 45; the middle one is 8; 8 must appear twice, and it can only be the 2nd number, so the 4th is 45 − 5 − 8 − 8 − 14 = 10)

**X1.** 12 (5 × 8 = 40, 4 × 7 = 28, 40 − 28 = 12)

**X2.** 50 (10 × 32 = 320, 9 × 30 = 270, 320 − 270 = 50)

**X3.** 15 (2, 2, 5, 6, 15: the total is 30, so the last two make 21; the 4th must be at least 6, or 5 would tie with 2 as a mode)

**X4.** Yes, e.g. 1, 1, 1, 1, 16 has mean 4, which is bigger than four of the numbers.

**X5.** It went up by 12 (the total went from 60 to 72)

**X6.** 24 ((7 + 5) × 2)

**X7.** 3 sets: 1, 4, 7; 2, 4, 6; 3, 4, 5 (the total is 12, the middle is 4, so the other two add to 8 and one is less than 4)

**X8.** Pair them: 1 + 99, 2 + 98, … , 49 + 51 each make 100, with 50 left in the middle. Every pair balances around 50, so the mean is 50 (total 4950 ÷ 99 = 50). Program: set total to 0, set n to 1, repeat 99 (change total by n, change n by 1), say total ÷ 99.

**X9.** 68 kg (9 × 44 = 396, 8 × 41 = 328, 396 − 328 = 68)

## Lesson 10: Showcase Challenge

**C1.** 62 (nth term 3n + 2)

**C2.** 2 × 2 × 3 × 5

**C3.** −8 °C

**C4.** 14

**C5.** 6 (35, 37, 53, 57, 73, 75)

**C6.** 14 (70 ÷ 5)

**T1.** Pattern 33 (3n + 1 = 100)

**T2.** 83, 89 and 97

**T3.** 12 (the turn at each corner is 180° − 150° = 30°, and 360 ÷ 30 = 12)

**T4.** 6/36 = 1/6 (10: 3 ways, 11: 2 ways, 12: 1 way)

**T5.** RFYMX

**T6.** 12 o'clock (10 + 50 = 60, and 60 is a multiple of 12)

**T7.** 16 (7 × 10 − 6 × 9)

**T8.** 24 (4 × 3 × 2 × 1)

**X1.** 33 (multiples of 2: 50, of 3: 33, of both (6): 16; 100 − (50 + 33 − 16) = 33)

**X2.** 4 (4 × 3 − 8 = 4). With + 8 it is −4 (−4 × 3 + 8 = −4).

**X3.** 30 (16 + 9 + 4 + 1)

**X4.** 12 (1, 2, 3, 4, 6, 12)

**X5.** 2 (5 × 144° = 720° = 2 × 360°)

**X6.** 4, 7 and 13 (the total is 24, so the smallest and biggest add to 17 and differ by 9)

**X7.** 4 ways: 3 + 47, 7 + 43, 13 + 37, 19 + 31

**X8.** 8/64 = 1/8 (1 + 8, 2 + 7, … , 8 + 1), so about 125 in 1000. Program: set nines to 0; repeat 1000 (if pick random 1 to 8 + pick random 1 to 8 = 9 then change nines by 1); say nines.

**X9.** Yes: it is the 338th term (6 × 338 − 1 = 2027). The first term bigger than 2027 is the 339th, 2033.